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Snowflake

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Snowflake

Below is an example of using [Snowflake ↗](#) to connect and query data from R2 Data Catalog (read-only).

Prerequisites

- Sign up for a [Cloudflare account ↗](#).
- [Create an R2 bucket](#) and [enable the data catalog](#).
- [Create an R2 API token](#) with both [R2 and data catalog permissions](#).
- A [Snowflake ↗](#) account with the necessary privileges to create external volumes and catalog

integrations.

Example usage

In your Snowflake [SQL worksheet ↗](#) or [notebook ↗](#), run the following commands:

```
-- Create a database (if you don't already
have one) to organize your external data
CREATE DATABASE IF NOT EXISTS r2_example_db;

-- Create an external volume pointing to your
R2 bucket
CREATE OR REPLACE EXTERNAL VOLUME ext_vol_r2
    STORAGE_LOCATIONS = (
        (
            NAME = 'my_r2_storage_location'
            STORAGE_PROVIDER = 'S3COMPAT'
            STORAGE_BASE_URL =
's3compat://<bucket-name>'
            CREDENTIALS = (
                AWS_KEY_ID = '<access_key>'
                AWS_SECRET_KEY =
'<secret_access_key>'
            )
            STORAGE_ENDPOINT =
'<account_id>.r2.cloudflarestorage.com'
        )
    )
    ALLOW_WRITES = FALSE;
```

-- Create a catalog integration for R2 Data Catalog (read-only)

CREATE OR REPLACE CATALOG INTEGRATION

r2_data_catalog

CATALOG_SOURCE = ICEBERG_REST

TABLE_FORMAT = ICEBERG

CATALOG_NAMESPACE = 'default'

REST_CONFIG = (

CATALOG_URI = '<catalog_uri>'

CATALOG_NAME = '<warehouse_name>'

)

REST_AUTHENTICATION = (

TYPE = BEARER

BEARER_TOKEN = '<token>'

)

ENABLED = TRUE;

-- Create an Apache Iceberg table in your selected Snowflake database

CREATE ICEBERG TABLE my_iceberg_table

CATALOG = 'r2_data_catalog'

EXTERNAL_VOLUME = 'ext_vol_r2'

CATALOG_TABLE_NAME = 'my_table'; *-- Name*

of existing table in your R2 data catalog

-- *Query your Iceberg table*

```
SELECT * FROM my_iceberg_table;
```

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